1. What Are We Predicting?

We want to predict whether a customer who buys auto insurance also has a home insurance policy with the same company. This is called a "bundled" account. Knowing who is likely to bundle helps the marketing team reach out to the right customers for cross-selling.

Item	Details
Target variable	POL_ACCT_CR_IND
What 0 means	Customer has auto insurance only
What 1 means	Customer has both auto and home insurance (bundled)
Type of problem	Binary classification (yes or no)
Each row represents	One auto insurance policy (300,000 total rows)
When we predict	At the time the customer gets a quote or renews their auto policy

Why PR-AUC as our main metric?

The bundled customers are the minority group (roughly 1 in 4). We care most about correctly identifying them at the top of our list. PR-AUC (Average Precision) is the right metric here because:

- **Focus:** It focuses on how well we rank the actual bundled customers at the top.
- **Robustness:** It is not fooled by the large number of auto-only customers (unlike ROC-AUC, which can look good even with poor minority-class performance).
- **Flexibility:** It evaluates performance across all possible thresholds, so we are not locked into one cutoff.

2. Dataset Overview

Property	Value
Number of rows	300,000 policy records
Number of columns	44 features
File size	~150 MB
Special loading note	Used keep_default_na=False so the string NA in RISK_FLAG is not confused with missing data

Data Quality Issues Found

During the data audit, we found and fixed several issues:

- **Row ID:** AUTO_POL_TRANS_ID is just a row number — it would let the model memorize answers instead of learning real patterns. Dropped.
- **Constant column:** AARP_MEMB_IND has the same value (Y) for every row, so it gives zero useful information. Dropped.
- **String NA trap:** RISK_FLAG looks like missing data, but NA is actually a real category. We preserved it with special loading, but since >95% of values are NA, it has almost no predictive power. Dropped.
- **Leakage column:** POL_FORM_CD contains home insurance codes (HO3, HO4, HO6) — this directly reveals the answer we are trying to predict. This is data leakage. Dropped.
- **Excel artifact:** AQD_GRP has a label 13-Jul that was created when Excel wrongly converted the range 7-13 into a date. Recoded back to 7-13.

3. Missing Data & How We Handle It

Most columns have no missing values. A few have some, and we chose an imputation strategy for each:

Column	% Missing	Why it is missing	What we do
CLTV_LONGVTY_GRP_CD	~15–20%	New customers do not have a longevity score yet	Add an Unknown category (missingness itself is useful info)

Car price columns (MIN/MAX/AVG)	~2–5%	Incomplete vehicle lookup in the quoting system	Fill with the median value (fit on training data only)
MKT_CHNNL_NM, MKTG_MEDIA_GRP_DESC	~5–10%	Some channels do not record marketing attribution	Add an Unknown category
PCARR_CALC_YR_CNT	~3–5%	First-time buyers have no prior carrier history	Fill with the median value
ADV_QTE_DAY_CNT	<1%	Random — no pattern found	Fill with the median value

Important note: For CLTV_LONGVTY_GRP_CD, we assume the data is Missing At Random (MAR), meaning the missingness depends on things we can observe (like whether the customer is new). We cannot fully rule out that missingness depends on the score itself (MNAR). If this feature becomes very important in the model, we recommend a sensitivity analysis.

4. Target Variable Analysis

Class Balance

The target is moderately imbalanced — roughly 3 to 4 auto-only customers for every 1 bundled customer. With 300,000 rows, even the minority class has tens of thousands of examples, which is plenty for reliable model training.

How we handle the imbalance

- **Class weights:** Use class_weight='balanced' in the model so it pays more attention to the minority class.
- **Threshold tuning:** After training, tune the decision threshold on the validation set instead of using the default 0.5.
- **No resampling needed:** With 300K rows we have enough data — SMOTE or undersampling would add noise without benefit.

Stability over time

The bundling rate stays steady across all months in the data — no trends, no seasonal patterns, no sudden shifts. This is good news because it means:

- We can use a simple random split (no need for a time-based split).
- We should stratify the split by the target so each partition keeps the same class ratio.
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5. Top-10 Strongest Predictors

We identified the best features using three independent methods: bivariate statistics (Stage 7), mutual information (Stage 9a), and random forest importance (Stage 9c). Features that rank high across all three are the most trustworthy:

Rank	Feature	Why it predicts bundling
1	BILL_PYMENT_FREQ_DESC	Customers who pay annually tend to be homeowners with stable finances
2	POL_BI_OCCUR_LMT_CD	Higher liability limits suggest the customer owns assets worth protecting
3	HH_COMP	Household type (e.g., married couple) strongly affects homeownership likelihood
4	AVG_VEH_PREM	Higher vehicle premiums suggest wealthier households more likely to own homes
5	RISK_ST_ABBR	Homeownership rates vary a lot by state
6	PCARR_DESC	Customers from multi-line carriers (like State Farm) are already used to bundling
7	POL_MULTI_SNGL_VEH_CD	Multi-vehicle households are larger and more likely to own a home
8	ORIG_QTE_CNTCT_METH_CD	How the customer got a quote (online vs. phone) relates to their

		demographics
9	AQD_GRP	How far in advance they requested a quote shows planning behavior
10	RISK_GRP	The underwriting risk tier relates to policyholder stability

6. Leakage Check

Data leakage means a feature contains information that would not be available when making a real prediction. If we accidentally include such a feature, the model looks great in testing but fails completely in production.

Confirmed leakage: POL_FORM_CD — DROPPED

This column contains home insurance policy codes like HO3, HO4, and HO6. By definition, if someone has a home policy code, they are bundled. This feature directly reveals the answer. It had near-perfect prediction power, which is a red flag. It was removed from the dataset.

Under monitoring: BILL_PYMENT_FREQ_DESC — KEPT (with caution)

Payment frequency is chosen when the customer signs their policy, so it is technically available at prediction time. However, there is a possible concern: the company may offer annual-pay discounts specifically to bundled customers. If that is the case, the feature partly reflects the outcome rather than predicting it. We recommend running an ablation test (train the model with and without this feature) to check whether it inflates metrics unfairly.

7. Known Risks

These are potential issues to be aware of as we move into modeling:

7.1 Multicollinearity (redundant features)

Some features measure very similar things. Keeping all of them wastes model capacity and can make linear model coefficients unreliable.

What we found	Action taken
POL_TTL_BILL_PREM_AMT is nearly equal to AVG_VEH_PREM x PPV_CNT (VIF > 10)	Dropped POL_TTL_BILL_PREM_AMT; kept AVG_VEH_PREM and PPV_CNT separately
PLCY_MIN, MAX, and AVG car base price are nearly identical	Dropped MIN and MAX; kept AVG only
POL_RATE_DRVR_MIN_AGE and MAX_AGE are correlated	Kept both — they carry different info for multi-driver households

7.2 Fairness Concerns

Some features may act as proxies for protected characteristics:

- RISK_ST_ABBR (state) correlates with racial and socioeconomic demographics.
- Driver age features are a protected class in some regulations.
- HH_COMP (household composition) may relate to marital status.

Recommendation: Run a disparate-impact analysis (80% rule) and equalized-odds check across age groups, states, and household types before deploying the model.

7.3 Data Drift

The target rate is stable right now, but things can change after the model is deployed. Marketing strategies, competitor behavior, and economic conditions can shift the patterns the model learned.

Recommendation: Monitor feature distributions (flag if PSI > 0.25) and track the monthly bundling rate. If it drifts more than 2 standard deviations from the training period, retrain the model.

8. Train / Validation / Test Split

Partition	Percentage	~Rows	Purpose
Training	70%	~210,000	The model learns from this data
Validation	15%	~45,000	Used to tune settings and pick the best threshold
Test	15%	~45,000	Used once at the end to report final performance — never used for decisions

Why this split?

- **No time dependency:** The target rate is stable over time, so a random split is appropriate (no need for a time-based split).
- **Stratified:** We stratify by the target variable so each partition has the same ratio of bundled vs. auto-only.
- **Large enough:** Each evaluation partition has ~11,000–13,000 positive cases, which is enough for reliable PR-AUC estimates.

Code

```
X_train, X_temp, y_train, y_temp = train_test_split(
    X, y, test_size=0.30, stratify=y, random_state=42)
X_val, X_test, y_val, y_test = train_test_split(
    X_temp, y_temp, test_size=0.50, stratify=y_temp, random_state=42)
```

9. Preprocessing Pipeline

Golden rule: All transformations are fit on the training set only, then applied to validation and test sets. This prevents information leakage.

Feature Type	Examples	What We Do
Skewed numbers	AVG_VEH_PREM, RENEW_YRS, ADV_QTE_DAY_CNT, car price	Fill missing with median, then apply Yeo-Johnson transform (handles zeros, finds best power automatically)
Normal numbers	Driver ages, AARP_MEMB_YR_CNT	No imputation needed, just StandardScaler
Count	PPV_CNT (number of vehicles)	StandardScaler only
Ordered categories	RISK_GRP, POL_BI_OCCUR_LMT_CD, AQD_GRP, CLTV_LONGVTY_GRP_CD	OrdinalEncoder with the correct tier order; Unknown for missing CLTV; fix 13-Jul to 7-13
Binary (0/1 numbers)	FULL_PAY_DISC_IND, E_SIGN_IND	Keep as-is (already encoded)
Binary (text)	POL_NEW_RENEW_CD, CLEAN_DIRTY_DESC	OrdinalEncoder (maps 2 text levels to 0/1)
Categories (few levels)	BILL_PYMENT_FREQ_DESC, BILL_PYMNT_METH_DESC	OneHotEncoder (handle_unknown='ignore')
Categories (many levels)	RISK_ST_ABBR, PCARR_DESC, HH_COMP	Group rare levels (<1%) into Other, then OneHotEncoder

Marketing columns	MKT_CHNNL_NM, MKTG_MEDIA_GRP_DESC	Fill missing with Unknown, then OneHotEncoder
Time features	POL_EFF_MONTH, POL_EFF_YEAR	OneHotEncoder or passthrough

Features Dropped Before Modeling

Feature	Why it was dropped
AUTO_POL_TRANS_ID	Row ID — the model would memorize it
POL_FORM_CD	Leakage — it reveals the answer
AARP_MEMB_IND	Constant — same value for everyone
RISK_FLAG	Nearly constant — >95% the same value
POL_EFF_DT, POL_EXP_DT	Raw dates replaced by month/year features
POL_TTL_BILL_PREM_AMT	Redundant with AVG_VEH_PREM (VIF > 10)
PLCY_MIN/MAX_CAR_BASE_PRICE	Redundant with AVG car price (VIF > 10)

10. Open Questions for Modeling Phase

These are questions we cannot fully answer during EDA. They should be addressed during the modeling stage:

1. Is BILL_PYMENT_FREQ_DESC safe to use? It is the strongest predictor, but it may partly reflect the bundling decision itself (if bundled customers get special annual-pay discounts). An ablation study — training the model with and without this feature — will tell us how much the metrics change.

2. How should we encode RISK_ST_ABBR? With 50 states, one-hot encoding creates many columns. Target encoding (replacing each state with its average bundling rate) is more compact but can overfit. A cross-validation comparison is needed to decide.

3. Is the CLTV_LONGVTY_GRP_CD missingness truly random? We assume it is missing because of customer tenure (MAR), but we cannot prove it. If this feature becomes important in the model, we should test whether the results change under different missingness assumptions.

4. Is the model fair? Before deploying, we need to check whether the model treats different demographic groups fairly. This includes checking for disparate impact across age groups, states, and household types.

5. How do we monitor the model in production? We need to define thresholds for when the model should be retrained — for example, if feature distributions shift significantly (PSI > 0.25) or the monthly bundling rate drifts outside its historical range.